LOVENTA ANYANGO OTIENO

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PROFESSIONAL SUMMARY

Geomatic Engineering graduate (JKUAT, 2025) with hands-on experience in GIS, remote sensing, and land surveying, and a track record of applying Python-based geospatial analysis to real-world environmental and infrastructure projects. Published co-author of two peer-reviewed journal articles on remote sensing and machine learning for forest ecosystem monitoring, and a book chapter on the effects of climate change on crop production. Skilled at translating field survey and satellite data into actionable maps and reports; recognized for collaborative, solutions-driven delivery of high-quality geospatial outputs.

CORE SKILLS

GIS & Remote Sensing: ArcGIS, QGIS, Google Earth Engine, ERDAS Imagine, LiDAR

Programming & Data Analysis: Python, R, JavaScript, PostGIS, Machine Learning, Deep Learning

Surveying & CAD: AutoCAD, Civil 3D, Total Station, RTK, Cadastral Surveying

Cloud & Data Platforms: Oracle Cloud Infrastructure, Copernicus, Earthdata, USGS EarthExplorer

Core Competencies: Communication, Critical Thinking, Research, Teamwork, Problem-Solving

PROFESSIONAL EXPERIENCE

Geospatial Intern, Nile Surveys and Geo-Solutions Ltd Dec 2025 – Nov 2026

- Digitized drone imagery to support mapping and analysis workflows.
- Created maps for multiple client projects.
- Performed static field observations and beacon re-establishment using RTK and Total Station equipment.
- Conducted engineering road surveys using Total Stations.
- Prepared field survey reports for the Affordable Housing Program across select project areas.

Geospatial Intern, Kenya Space Agency Jan 2024 – Apr 2024

- Developed Python scripts to process satellite imagery and visualize remote sensing outputs for agricultural monitoring under the Agriculture Information Management System for Murang'a Project.
- Reviewed and edited Space Club educational content to improve scientific accuracy and readability for young audiences.
- Served as rapporteur for the Earth Observation Workshop, compiling and presenting a comprehensive final report.
- Coordinated data entry and logistics for Kenya Space Expo 2024, supporting smooth event delivery.

Survey Intern, Ministry of Lands, Public Works, and Urban Planning – Thika Jan 2023 – Mar 2023

- Assisted in cadastral surveys and field documentation.
- Conducted mutation checks and computations to validate land parcel changes.

Relationship Officer, Equity Bank Kenya Ltd – Westlands Feb 2019 – Aug 2019

- Delivered customer service support and handled teller duties, transaction processing, and financial record maintenance.

EDUCATION

B.Sc. in Geomatic Engineering and Geospatial Information Systems Sept 2019 – June 2025
Jomo Kenyatta University of Agriculture and Technology (JKUAT), Juja

Kenya Certificate of Secondary Education (KCSE) – Grade A (82 points) Feb 2015 – Nov 2018
Moi Forces Academy, Lanet

PUBLICATIONS

- Kipkulei, H., Awandu, W., Otieno, T. A., Otieno, L. A., Ngoya, Z. J., & Rotich, B. (2027). Consequences of climate change on crop production. In T. M. Letcher (Ed.), Climate change (pp. 239–259). Elsevier. https://doi.org/10.1016/B978-0-443-32834-3.00010-4
- Otieno, L. A., Otieno, T. A., Rotich, B., Löhr, K., & Kipkulei, H. K. (2025). Integrating remote sensing and machine learning to evaluate environmental drivers of post-fire vegetation recovery in the Mount Kenya forest. Discover Geoscience, 3(1), 81. https://doi.org/10.1007/s44288-025-00196-5
- Otieno, T. A., Otieno, L. A., Rotich, B., Löhr, K., & Kipkulei, H. K. (2025). Modeling climate change impacts and predicting future vulnerability in the Mount Kenya forest ecosystem using remote sensing and machine learning. Environmental Monitoring and Assessment, 197(6), 631. https://doi.org/10.1007/s10661-025-14089-0

CERTIFICATIONS

- Python for Data Science, AI & Development — IBM (Credential ID: N1XL9G0PFKFA)
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle (Testing ID: OC6664691; Aug 2025 – Aug 2027)

PROFESSIONAL MEMBERSHIP

Institution of Surveyors of Kenya (ISK) — Membership No. 7081

LANGUAGES

English (Fluent) | Swahili (Fluent) | Luo (Fluent) | French (Intermediate)

REFEREES

Dr. rer. agr. Harison Kipkulei — Lecturer, Department of Geomatic Engineering, JKUAT | hkipkulei@jkuat.ac.ke | 0717 623 982

Mr. Conrad Olimba — Surveyor, Nile Surveys & Geo-Solutions Ltd | 0703 305 553